Domain-Specific Fine-Tuning for Reliability Engineering: From Small-Seed Self-Instruct to Reinforcement Learning

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Abstract

Large language models show strong general reasoning but struggle with domain-specific technical tasks such as reliability engineering, where answers require precise numerical computation and specialized knowledge. We present a complete pipeline for adapting a general-purpose LLM to the reliability engineering domain, starting from only 56 seed questions extracted from textbooks. We introduce an adapted self-instruct methodology that replaces same-model answer consistency with cross-model verification, combined with paraphrase-based augmentation inspired by MetaMath, to scale the dataset to 866 high-quality question-reasoning-answer triplets. We identify the self-instruct ceiling effect, wherein LLM generators have a tendancy to produce questions systematically easier than real domain problems. We show that paraphrase augmentation is significantly more effective than difficulty scaling for overcoming it. Supervised fine-tuning of Qwen3-8B with QLoRA yields a statistically significant $+18.6\%$ *accuracy improvement* ($p = 0.002$) *under 10-fold cross-validation, placing our 8B-parameter model between frontier models o3-mini and Gemini 3.1 Pro on reliability engineering tasks. We further apply Group Relative Policy Optimization (GRPO) to improve reasoning quality beyond what SFT alone achieves. Our results demonstrate that small-seed, LLM-driven data synthesis pipelines can produce effective domain-specific training data, and that combining SFT with reinforcement learning from verifiable rewards is a promising approach for low-resource technical domains.*

[TODO: Refine after all results are in.]

1. Introduction

Reliability engineering is concerned with the analysis and prediction of failure in complex systems—from nuclear power plants and aircraft to electrical grids and autonomous vehicles. Practitioners routinely employ sophisticated mathematical tools including Weibull and exponential lifetime distributions, Markov chain modeling, Bayesian inference, fault tree analysis, and system-level reliability computation for series-parallel and k-out-of-n architectures. Mastering these techniques requires years of specialized training, and even experienced engineers benefit from computational assistance when solving multi-step quantitative problems.

Large language models (LLMs) have demonstrated impressive general reasoning capabilities across a wide range of tasks [9]. However, when applied to domain-specific technical problems such as reliability engineering, their performance degrades substantially. Prior work evaluated state-of-the-art models on a set of hand-written reliability engineering problems and found that even frontier models fail to produce correct numerical answers in the majority of cases, often generating plausible-sounding but mathematically incorrect solutions. This performance gap motivates the development of domain-adapted models that can reliably assist with reliability engineering computations.

A central challenge in building such models is the scarcity of training data. Unlike well-studied domains such as general mathematics [5, 24] or medicine [17], reliability engineering lacks large-scale question-answer datasets or established benchmarks. Manually creating such datasets is prohibitively expensive, as each problem requires expert knowledge to both formulate and solve. This creates a bootstrapping problem: we need data to train a model, but generating quality data requires the very expertise the model is meant to provide.

We address this challenge with a data generation and fine-tuning pipeline that bootstraps 56 textbook-extracted seed questions into 866 verified training examples, achieving $+18.6\%$ accuracy ($p = 0.002$) on Qwen3-8B. Our contributions are:

- A cross-model answer verification method for synthetic data quality control, replacing same-model consistency checks that we show are trivially satisfied in technical do-

*Code and data: https://github.com/ADnocap/Reliability-Domain-Specific-LLM

†Experiments (exp) described in appendix A

mains (Section 3.2).

- The identification and empirical characterization of the self-instruct ceiling effect (Section 3.2), and evidence that paraphrase augmentation overcomes it more effectively than difficulty scaling (Section 3.4).
- A reproducible SFT pipeline achieving $+18.6\%$ on reliability engineering with a fine-tuned 8B model competitive with frontier models an order of magnitude larger (Section 4).
- Integration of GRPO with verifiable rewards to further improve domain reasoning beyond SFT (Section 5).

2. Related Work

2.1. Domain-Specific LLM Adaptation

Fine-tuning LLMs for specialized domains has shown promising results across several fields. In medicine, Med-PaLM [17] achieved expert-level performance on medical question answering through instruction tuning on curated clinical datasets, while PMC-LLaMA [21] demonstrated that continued pre-training on biomedical literature followed by instruction tuning can substantially improve domain performance. In the legal domain, models fine-tuned on case law and statutory text have shown improved performance on legal reasoning tasks [1]. For code generation, Code Llama [14] showed that domain-specific continued pre-training followed by instruction tuning outperforms general-purpose models.

A common finding across these efforts is that domain adaptation is most effective when high-quality, domain-specific training data is available. However, for niche technical fields like reliability engineering where experts are proficient in multiple technical disciplines such as probability theory, statistics, and systems engineering, no such datasets or benchmarks currently exist. Our work addresses this gap by developing both a data generation pipeline and a fine-tuning methodology specifically designed for low-resource technical domains.

2.2. Self-Instruct and Synthetic Data Generation

Self-Instruct [20] introduced a framework for bootstrapping instruction-following data by using a language model to generate its own training examples from a small seed set of 175 human-written tasks. The method produced 52K instruction-output pairs, improving GPT-3's instruction following by $+33.1\%$ on Super-NaturalInstructions. To ensure diversity, generated instructions are only added to the task pool if their ROUGE-L similarity with all existing instructions falls below 0.7. However, this diversity filter says little about *correctness*, in fact, only 54% of generated examples were rated as fully valid, with output quality identified as the weakest component.

Several works have extended this framework. Stanford Alpaca [18] scaled self-instruct to generate 52K examples using GPT-3.5 for fine-tuning LLaMA. WizardLM [22] introduced Evol-Instruct, which iteratively increases instruction complexity through depth and breadth evolution. Ranaldi and Freitas [13] proposed Self-Refine Instruction-Tuning, a two-stage approach where LLMs first generate chain-of-thought demonstrations for smaller student models, which then self-refine via DPO.

For verifiable tasks such as mathematics, answer-consistency filtering has been proposed as a quality control mechanism [23]. The idea is straightforward: after generating a question-answer pair, the same model is prompted to solve the question independently a n times; if more than $n/2$ instances agree on the anwer, the example is considered reliable and retained, otherwise it is discarded. This targets the correctness gap left by ROUGE-L, which only filters for diversity. However, a subtle issue arises when the same model is used for both generation and verification. We observe empirically that models can already solve the vast majority of questions they generate. In our domain, baseline models achieve 92–97% accuracy on self-instruct-generated questions while scoring only 68–76% on real textbook problems. This means same-model answer consistency is trivially satisfied and does not filter for difficulty or correctness in any meaningful way. While this does not invalidate the downstream fine-tuning results of these approaches (the trained model may still improve through exposure to diverse reasoning patterns), it reveals that answer consistency is not a reliable proxy for data quality in technical domains.

This observation is related to a broader limitation identified by SearchInstruct [8]: self-instruct "depends solely on internal knowledge," meaning the generator can only create questions within its own capability envelope. They propose grounding generation in retrieved documents to overcome this.

Our approach addresses the correctness problem through *cross-model* answer verification: an independent model (Claude Opus 4) generates questions from the seed dataset and solves them, we then use a comparably powerful model (ChatGPT 5.4) to verify the solution by solving each question without seegin the answer or reasoning trace, and only items where both models agree on the numerical answer (within 5% tolerance) are retained. Because the verifier is a different model with different failure modes, agreement provides a stronger signal than same-model consistency. We additionally use a stronger generator model than the fine-tuning target, and generate full chain-of-thought triplets (question, reasoning, answer) rather than instruction-output pairs.

2.3. Data Augmentation for Multi-Step Mathematical Reasoning

Data augmentation has emerged as a critical strategy for improving mathematical reasoning in LLMs. MetaMath [24] demonstrated that rephrasing mathematical questions from multiple perspectives (without changing the underlying problem) significantly improves reasoning performance, achieving state-of-the-art results on GSM8K and MATH benchmarks. The key insight is that surface-level diversity in question formulation is more valuable than increasing problem difficulty. PersonaMath [19] extended this idea by generating question variants from different persona perspectives.

Ding *et al*. [4] provide a comprehensive survey of LLM-based data augmentation, finding that while larger generator models produce higher-quality data, cross-model validation is essential to prevent self-reinforcing biases. Shumailov *et al*. [16] formalized this concern as "model collapse," showing that training on model-generated data without external grounding leads to progressive quality degradation across generations.

Our paraphrase augmentation strategy is directly inspired by MetaMath: we use Claude Opus 4.6 to rephrase questions while strictly preserving all numerical values and mathematical relationships, then verify meaning preservation with GPT-5.4. This approach scales our dataset from 280 to 866 examples and yields our strongest results ($+18.6\%$, $p = 0.002$), consistent with the finding that surface diversity outperforms difficulty scaling.

2.4. Reinforcement Learning for Reasoning

Reinforcement learning from human feedback (RLHF) [10] established the paradigm of aligning LLMs with human preferences through reward modeling and policy optimization. Direct Preference Optimization (DPO) [12] simplified this by eliminating the need for a separate reward model, directly optimizing the policy from preference pairs. However, both methods rely on human-generated or model-generated preference data, which can be noisy for technical domains.

For tasks with verifiable answers, such as mathematics, reinforcement learning from verifiable rewards offers a compelling alternative. DeepSeek-Math [15] introduced Group Relative Policy Optimization (GRPO), which generates multiple candidate responses per question, scores them against ground-truth answers, and computes group-relative advantages without requiring a critic model. DeepSeek-R1 [2] scaled this approach, demonstrating that GRPO can elicit sophisticated reasoning behaviors including self-verification and chain-of-thought refinement.

Reliability engineering problems are particularly well-suited for GRPO because they produce verifiable numerical answers that can serve as ground-truth rewards. In our pipeline, SFT first provides domain knowledge and reasoning format, after which GRPO optimizes the model to prefer reasoning chains that lead to correct final answers. Our early experiments with GRPO on small datasets (215 questions) showed limited improvement over SFT alone ($+0.5\%$ vs. $+2.3\%$), motivating the use of larger augmented datasets for the RL stage.

[TODO: Add more on GRPO with our results form ELORA.]

2.5. Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) [6] enables efficient fine-tuning by adding trainable low-rank matrices to frozen pre-trained weights, drastically reducing the number of trainable parameters. QLoRA [3] combines LoRA with 4-bit quantization, enabling fine-tuning of large models on consumer-grade hardware. NEFTune [7] demonstrated that adding uniform noise to embedding vectors during fine-tuning acts as an effective regularizer, improving instruction-following performance by 25–35% on standard benchmarks.

We employ 4-bit QLoRA with LoRA applied to all attention and MLP projection layers, combined with NEFTune ($\alpha = 5$) as a regularizer against overfitting on our small training sets (approximately 780 examples per fold in 10-fold cross-validation on 866 total examples).

3. Data Pipeline

Our data pipeline transforms 56 textbook-extracted seed questions into 866 verified training examples through four stages: seed extraction, synthetic generation with cross-model verification, cleaning, and paraphrase augmentation. Figure 1 illustrates the complete pipeline.

3.1. Seed Dataset Extraction

We digitized 32 chapters from reliability engineering textbooks using Mistral OCR, converting them to structured markdown. The textbooks cover exponential and Weibull lifetime distributions, mean time to failure (MTTF) and hazard functions, Bayesian inference, series-parallel and k-out-of-n system reliability, reliability demonstration testing, and acceptance sampling.

[TODO: need to quote and mention which textbook, I cant remember which it is and maybe we worl this part above]

From this corpus, we extracted question-reasoning-answer (QRA) triplets through a three-step process: (1) keyword scanning with regular expressions to identify exercise and solution blocks, (2) LLM-assisted validation to check self-containment (no references to figures or prior parts), and (3) structured extraction into (q, r, a) triplets using GPT-5-Mini. After iterative cleaning—including LaTeX normalization, Unicode fixing, and manual curation—

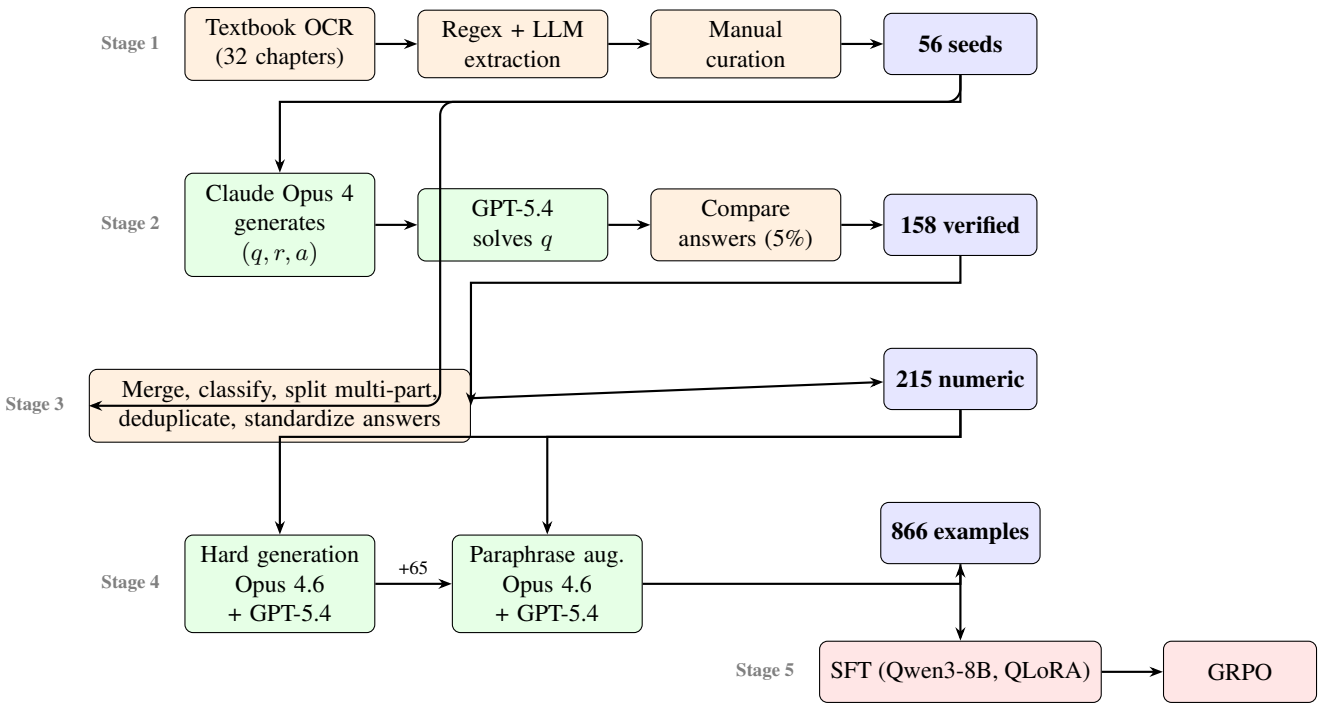

Figure 1. Data pipeline overview. Blue: dataset milestones; orange: processing; green: LLM-based generation/verification; red: training. Seed questions feed both the generation stage and the cleaning stage (routed via the left margin to avoid crossing Stage 2 nodes). **[TODO: FIX THE FUCKING ARROWS]**

we obtained **56 high-quality seed QRA triplets** covering the core topics of reliability engineering.

3.2. Adapted Self-Instruct with Cross-Model Verification

3.2.1. Generation with a Stronger Model

Unlike the original Self-Instruct [20], which uses the same model for both generation and fine-tuning, we employ *stronger* generator models to create training data for a smaller target model (Qwen3-8B). This asymmetry is motivated by the observation that the target model lacks the domain knowledge needed to generate correct reliability engineering problems and solutions.

Each generation prompt includes 3 randomly sampled seed examples and requests 2 new problems. The generator is instructed to produce complete (q, r, a) triplets with step-by-step chain-of-thought reasoning, using temperature 0.8 for diversity. Quality gates filter out questions shorter than 30 characters, reasoning shorter than 50 characters, placeholder answers, and near-duplicates (by 200-character question prefix). This process yielded 158 synthetic QRA triplets that passed initial quality checks.

3.2.2. Cross-Model Answer Verification

To ensure mathematical correctness, we replace same-model answer consistency [23] with a cross-model verification pipeline:

1. **Generate and solve:** Claude Opus 4 produces a (q, r, a) triplet from the seed pool (temperature 0.8).
2. **Verify:** GPT-5.4 independently solves the same question without seeing the generator's answer or reasoning (temperature 0.1).
3. **Compare:** Final numerical answers are compared with 5% relative tolerance, with LLM-based semantic comparison as fallback for edge cases.

Only items where both models agree (MATCH) are retained. A safety valve halts generation if the acceptance rate drops below 10% after 20 batches. Because the generator and verifier are different models with different architectures and failure modes, agreement provides a substantially stronger correctness signal than same-model consistency.

3.2.3. The Limitations of Same-Model Answer Consistency

A key motivation for cross-model verification is our empirical observation that same-model answer consistency, as used in prior work [23], is trivially satisfied for self-instruct-generated questions. Table 1 shows that baseline models achieve 92–97% accuracy on synthetic questions while scoring only 68–76% on the 56 textbook-extracted seed problems.

This means that if the generating model is also used for

Table 1. Accuracy of baseline models on textbook-extracted seed questions (56) vs. self-instruct-generated questions, revealing the systematic difficulty gap.

Model	Textbook (56)	Synthetic
Qwen3-8B	68.2%	92.4%
GPT-4-Mini	76.0%	97.1%

[TODO: Verify exact numbers from evaluation logs. idf these numbers chnage, you need to chnage the referacnes to this below and in the intro]

answer consistency verification, it will confirm nearly all of its own outputs, regardless of whether they represent genuinely challenging or educationally useful problems. The verification step thus filters for self-consistency rather than for quality or difficulty. This does not invalidate the downstream fine-tuning improvements reported in prior work as the trained model may still benefit from exposure to diverse reasoning patterns, but it reveals that answer consistency is not a reliable proxy for data quality in technical domains where the goal is to teach *new* knowledge rather than to align instruction-following behavior.

3.2.4. Self-Instruct Ceiling Effect

The accuracy gap in Table 1 reflects a more fundamental limitation that we term the *self-instruct ceiling effect*: LLM generators create questions within their own capability envelope. The generator avoids edge cases it does not understand (truncated distributions, competing risks), defaults to standard formulations matching its training distribution, and produces clean numerical answers that do not require iterative methods or lookup tables. Consequently, synthetic questions are systematically easier than real domain problems.

This has a direct implication for fine-tuning: training on self-instruct data may teach the model the *format* of domain-specific reasoning (step-by-step solutions, correct use of notation) without imparting the *knowledge* needed to solve genuinely hard problems. This ceiling effect motivates our paraphrase augmentation strategy (Section 3.4): rather than generating more synthetic questions, we create surface-level rephrases of real textbook problems, preserving their difficulty while increasing training set diversity.

3.3. Dataset Cleaning and Standardization

The combined seed and verified synthetic data (140 items) underwent LLM-assisted cleaning using Claude Opus 4.6 (temperature 0.1):

1. **Type classification:** Each item was labeled as SINGLE_NUMERIC, MULTI_PART_NUMERIC, FORMULA, DERIVATION, or CONCEPTUAL.
2. **Multi-part splitting:** 107 multi-part questions were split into fully self-contained entries, with each sub-question rewritten to include all necessary context (removing references such as "from Part (a)").
3. **Answer standardization:** Trailing periods, units, and commas were stripped; percentages retained as-is; LaTeX notation cleaned (e.g., `\frac{14}{33}` → `14/33`).
4. **Deduplication:** 173 near-duplicate entries removed by 200-character question prefix matching.

The resulting **master dataset** contains **256 items**: 215 numeric, 21 formula, 19 text, and 1 boolean. For all fine-tuning experiments, we use the 215 numeric questions as the base, since these permit automated evaluation via numerical comparison.

3.4. Paraphrase Augmentation

3.4.1. Augmentation Strategy

Inspired by MetaMath [24] and PersonaMath [19], we augment the dataset by generating surface-level rephrases of existing questions while strictly preserving mathematical content. Each original question is rephrased by Claude Opus 4.6 (temperature 0.7) with explicit constraints: all numerical values, formulas, and intermediate calculations must remain identical; only wording, sentence structure, and scenario framing may change (e.g., "plant" → "factory", "compute" → "determine"). Two manually curated few-shot examples guide the model.

Each rephrase is then verified by GPT-5.4 (temperature 0.1), which checks five dimensions: numerical values preserved, mathematical meaning preserved, constraints preserved, solution equivalence, and reasoning equivalence. A local gate additionally extracts key numbers from both versions and rejects the rephrase (with up to 2 retries) if any number is missing from the rephrased version.

3.4.2. Surface Diversity vs. Difficulty Scaling

We also experimented with generating harder questions using Claude Opus 4.6, requiring problems that combine multiple distributions, conditional probability, or system-level analysis. This produced 65 verified hard questions (cross-verified with GPT-5.4), forming dataset v2 (280 items).

The contrast in downstream performance is striking. Adding 65 hard questions to the 215-item base yielded only $+2.0\%$ improvement (Exp. 16), whereas adding 221 paraphrases of existing questions yielded $+6.2\%$ (Exp. 18), and scaling to 586 paraphrases yielded $+18.6\%$ (Exp. 24). This confirms the MetaMath finding: for small datasets, surface-level diversity in question formulation is more valuable than increasing problem difficulty.

3.5. Dataset Summary

Table 2 summarizes the dataset evolution. All versions contain numeric-only questions with chain-of-thought reasoning and verified numerical answers.

Table 2. Dataset versions and their composition. All entries are numeric QRA triplets with verified answers.

Version	File	Size	Source
v1 (cleaned)	`cleaned.jsonl`	215	98 textbook + 158 verified
v2 (+ hard)	`v2.jsonl`	280	v1 + 65 hard generated
v3 (+ para.)	`v3.jsonl`	501	v2 + 221 paraphrased
v4 (+ para.)	**`v4.jsonl`**	**866**	**v2 + 586 paraphrased**

Table 3. Base model comparison on 215 numeric questions (5-fold CV). A strong baseline is critical: Llama's low starting point leaves little room for SFT gains.

Model	Baseline	Fine-tuned	Δ	p
Llama 3.1 8B	37.5%	39.8%	+2.4	0.25
Qwen3-8B	**70.7%**	**73.0%**	**+2.3**	0.50

Table 4. SFT training configuration (best setup).

Parameter	Value
Learning rate	2×10^{-4}
LR scheduler	Cosine
Optimizer	AdamW 8-bit
Batch size	2 (grad. accum. 4, eff. 8)
Max epochs	8 (early stop $\rightarrow$ epoch 4)
Warmup steps	5
Weight decay	0.01
NEFTune α	5
Precision	BF16
Max sequence length	2048 tokens

Each entry follows a chat-format structure with three roles: a system prompt establishing the reliability engineering expert persona, a user message containing the question, and an assistant message containing the chain-of-thought reasoning followed by a clearly delimited final answer.

4. Supervised Fine-Tuning

4.1. Base Model Selection

We evaluated two open-source instruction-tuned models as base candidates: Llama 3.1 8B [14] and Qwen3-8B [11], both in 4-bit quantized form via Unsloth. Table 3 shows that Qwen3-8B achieves a substantially higher zero-shot baseline on our reliability engineering questions (70.7% vs. 37.5%), likely due to its stronger mathematical pre-training. Moreover, fine-tuning Llama 3.1 8B caused a regression on numeric questions (-1.4pp) while improving only format-dependent categories (formula $+28.6$pp, text $+15.8$pp), suggesting catastrophic forgetting of mathematical capabilities. We therefore selected Qwen3-8B as our base model for all subsequent experiments.

By analyzing the model's thinking traces, we observed that the base model uses these blocks as *scratch work*: it explores solution avenues, backtracks when an approach seems wrong, and iterates toward the answer, much like a human solving a problem for the first time. Our SFT training data, by contrast, consists of polished step-by-step solutions that proceed directly to the correct answer without false starts. Training on these clean reasoning chains with thinking enabled would effectively require the model to "already know the answer" and go straight to a proof, conflicting with the exploratory nature of the thinking trace. We found that this mismatch caused the model to overfit, producing empty thinking tags followed by shallow pattern-matched answers. Disabling thinking during both training and inference resolved this: the model generates direct chain-of-thought responses consistent with the training format, while preserving its internal reasoning capabilities for use during GRPO (Section 5).

4.2. Training Setup

4.2.1. LoRA Configuration

We use 4-bit QLoRA [3] via the Unsloth framework, which provides optimized CUDA kernels for approximately 40% faster training. LoRA adapters [6] are applied to all attention and MLP projection layers. Key hyperparameters are rank $r = 16$, scaling factor $\alpha = 32$, and dropout 0.05.

4.2.2. Training Hyperparameters

Table 4 summarizes the training configuration. NEFTune [7] adds uniform noise to embedding vectors during training, serving as a regularizer against memorization on small datasets. We found $\alpha = 5$ to be optimal: $\alpha = 7$ degraded performance by -1.3pp (Exp. 17), and $\alpha = 10$ showed only marginal gains (Exp. 4). Early stopping with patience 2 (monitoring validation loss on a 10% held-out split) consistently converged to epoch 4 regardless of the maximum epoch setting (6 or 8), indicating a stable training dynamic.

4.2.3. Data Format

Training examples follow a three-turn chat format: a *system* message establishing the reliability engineering expert persona ("You are an expert in reliability engineering... Provide clear, accurate answers with step-by-step reasoning. At the end, clearly state your final answer after 'Final Answer:' "), a *user* message containing the question, and an *assistant* message containing the full chain-of-thought reasoning followed by the final answer. The tokenizer's `apply_chat_template()` handles model-specific formatting.

4.3. Evaluation Protocol

4.3.1. K-Fold Cross-Validation

Given the small dataset sizes (215–866 examples), we use k-fold cross-validation rather than a single train/test split to obtain reliable performance estimates. Initial experiments used 5-fold CV (172 train / 43 test per fold), but 5 folds limit the minimum achievable p-value to 0.0625 under the Wilcoxon test, precluding statistical significance at $p < 0.05$. We therefore moved to 10-fold CV for experiments on 866 samples (∼780 train / ∼87 test per fold), which achieves $p = 0.002$ when all folds show improvement.

For each fold, a fresh base model is loaded and fine-tuned from scratch to avoid information leakage between folds. All evaluation uses greedy decoding (`do_sample=False`) with `max_new_tokens=4096`, ensuring deterministic and reproducible results. Early experiments with stochastic decoding inflated improvements (e.g., $+7.0\%$ stochastic vs. $+2.3\%$ greedy in Exp. 3 vs. Exp. 12).

4.3.2. Answer Comparison

We extract the final answer from model responses using pattern matching ("Final Answer:", "The answer is:", "Therefore,") with fallback to the last non-empty line. Extracted answers are then compared against ground truth through a multi-level pipeline:

1. **Exact match:** Normalized strings (lowercase, strip whitespace, remove LaTeX wrappers such as `\boxed{}`, convert `\frac{a}{b}` to `a/b`).
2. **Numerical match:** Extract all numbers via regex; accept if all values match within 5% relative tolerance.
3. **Fraction-to-decimal:** Convert fractions (e.g., 14/33 ↔ 0.4242).
4. **Percentage conversion:** Normalize percentages (e.g., 25.5% ↔ 0.255).
5. **Partial match:** Substring containment for answers longer than 3 characters.

An answer is considered correct if any of the above criteria is satisfied.

4.3.3. Statistical Testing

We use the Wilcoxon signed-rank test to compare per-fold baseline and fine-tuned accuracies. This non-parametric paired test is appropriate because: (1) we have a small number of paired observations (5 or 10 folds), (2) we cannot assume normality of accuracy differences, and (3) the test is robust to outlier folds. We use $p < 0.05$ as the significance threshold.

We additionally track *question flips* wherein a question changes correctness between baseline and fine-tuned model, aggregated across all folds. A successful fine-tuning should show substantially more wrong→right flips than right→wrong regressions.

Table 5. SFT results across dataset sizes. Best configuration per dataset shown. All use Qwen3-8B, greedy decoding, LR=2×10^{-4}, NEFTune=5.

N	Folds	Ep.	Base	FT	Δ	p
215	5	4	70.7	73.0	+2.3	0.50
280	5	4	71.0	73.0	+2.0	0.69
501	5	4	59.1	65.3	+6.2	0.063
866	**10**	**8→4**	**63.6**	**82.2**	**+18.6**	**0.002**

Table 6. Ablation studies on 215 numeric questions (5-fold CV, greedy). Bold marks the best configuration.

Variation	Δ	p
Best (LR=2×10^{-4}, NEFTune=5, r=16, 4ep)	**+2.3**	0.50
Lower LR (1×10^{-4})	−0.5	1.0
Higher NEFTune ($\alpha = 7$)	−1.3	0.38
Lower LoRA rank ($r = 8$, $\alpha = 16$)	+1.4	0.88
3 epochs (vs. 4)	+0.9	0.88
5 epochs (vs. 4)	+1.9	0.50
DPO (instead of SFT)	−1.7	0.63
SFT + GRPO	+0.5	0.88

4.4. SFT Results

Table 5 presents the main SFT results across dataset sizes. The improvement scales consistently with dataset size: from $+2.3\%$ (not significant) on 215 original questions to $+18.6\%$ ($p = 0.002$) on 866 augmented questions. All 10 folds showed improvement in Exp. 24, with per-fold deltas ranging from $+12.6\%$ to $+26.4\%$.

The question flip analysis for Exp. 24 shows 206 wrong→right flips against only 45 right→wrong regressions (ratio 4.6:1), indicating that fine-tuning broadly improves performance rather than trading one type of error for another. By contrast, Exp. 12 (215 samples) showed only 23 wrong→right vs. 18 right→wrong (ratio 1.3:1), confirming that larger augmented datasets produce more decisive improvements.

Note that the baseline accuracy is lower on larger datasets (63.6% on 866 vs. 70.7% on 215) because the augmented test sets include paraphrased questions the base model has not seen. This makes the improvement more impressive: the fine-tuned model handles both original and paraphrased formulations effectively.

4.5. Ablation Studies

Table 6 summarizes key ablation results on the 215-question dataset (5-fold CV, greedy decoding).

Early stopping. Across all experiments with maximum epochs ≥ 6, early stopping (patience 2, monitoring validation loss) consistently selects epoch 4 as the best checkpoint. This holds for both 501-sample (Exp. 19) and 866-sample (Exp. 22, 24) experiments, suggesting a stable convergence point for this model and domain.

Learning rate. LR=2×10^{-4} consistently outperforms LR=1×10^{-4}, consistent with the recommendation that LoRA adapters benefit from learning rates approximately $10\times$ higher than full fine-tuning.

RL methods on small data. Both DPO (-1.7%) and SFT+GRPO ($+0.5\%$) underperformed pure SFT ($+2.3\%$) on 215 questions. DPO actually degraded performance, likely because the small dataset produced too few informative preference pairs. GRPO showed marginal gains but insufficient to justify the additional training complexity at this scale.

[TODO: Chnage this if needed with ur results ELORA]

5. Reinforcement Learning with GRPO

5.1. Motivation

[TODO: Why RL after SFT? Limitations of SFT alone (memorization vs. reasoning). GRPO as a natural next step.]

5.2. GRPO Overview

[TODO: Brief description of GRPO algorithm. Key advantage over PPO/DPO: no reward model needed for verifiable tasks.]

5.3. Reward Design

[TODO: Define reward function: correctness reward (numerical comparison) + format reward (reasoning structure).]

5.4. Training Configuration

[TODO: GRPO-specific hyperparameters: group size, KL penalty, learning rate schedule, number of iterations.]

5.5. Results

[TODO: Accuracy comparison. Qualitative analysis of reasoning improvements. Statistical significance.]

6. Experiments and Analysis

6.1. Comparison with Frontier Models

To contextualize our fine-tuned 8B model's performance, we evaluate three frontier models on two benchmarks: (1) a 500-question sample from our training dataset (v4), and (2) a fully independent 54-question holdout set composed of textbook problems, professor-authored exam questions, and French reliability exercises (*TD fiabilité*), none of which appear in the training data. For all models, we use the same answer extraction and comparison pipeline described in Section 4.3. When the extraction pipeline fails to parse a model's response (UNABLE_TO_EXTRACT), we exclude that question from the denominator rather than counting it as incorrect, to avoid penalizing models for formatting differences.

Table 7. Frontier model accuracy on our reliability engineering benchmarks. N_{eval} is the number of questions with successfully extracted answers.

Model	v4 500-sample		
	Correct	N_{eval}	Accuracy
Claude Sonnet 4.6	462	500	**92.4%**
o3-mini	427	494	86.4%
Gemini 3.1 Pro Preview	308	403	76.4%
Qwen3-8B (base)	63.6% (10-fold CV)		
Qwen3-8B (ours)	**82.2%** (10-fold CV)		

Table 8. Independent holdout evaluation (54 questions, no overlap with training data). Our fine-tuned 8B model results are pending.

Model	Holdout (54 questions)		
	Correct	N_{eval}	Accuracy
Claude Sonnet 4.6	43	53	**81.1%**
o3-mini	37	48	77.1%
Gemini 3.1 Pro Preview	19	23	82.6%
Qwen3-8B (base)	**[TODO: Pending]**		
Qwen3-8B (ours)	**[TODO: Pending]**		

On the v4 sample (Table 7), Claude Sonnet 4.6 leads at 92.4%, followed by o3-mini (86.4%) and Gemini 3.1 Pro (76.4%). Our fine-tuned Qwen3-8B achieves 82.2% under cross-validation on the same dataset, placing it between o3-mini and Gemini 3.1 Pro despite being an order of magnitude smaller. Note that the fine-tuned model's accuracy is measured via 10-fold CV (training on 90%, testing on 10%) and is therefore not directly comparable to the frontier models which see all questions at test time; the comparison is nevertheless informative as an approximate ranking.

On the independent holdout (Table 8), Claude Sonnet 4.6 achieves 81.1% and o3-mini 77.1%. Gemini 3.1 Pro achieves 82.6% on the 23 questions where answer extraction succeeded, though the high extraction failure rate (31/54 = 57.4%) limits the reliability of this estimate. Our fine-tuned model's holdout results are pending.

[TODO: Finish/chnage this when eval jobs are done]

Table 9. SFT improvement and question flips across dataset sizes. W→R: wrong→right, R→W: right→wrong.

N	Δ	p	W→R	R→W	Ratio
215	+2.3	0.50	23	18	1.3
280	+2.0	0.69	19	15	1.3
501	+6.2	0.063	74	43	1.7
866	**+18.6**	**0.002**	**206**	**45**	**4.6**

Table 10. Effect of training epochs on 866-question dataset (10-fold CV).

Epochs	FT Accuracy	Δ	p
3	75.1%	+11.4	0.002
4	79.3%	+15.7	0.002
6 (ES→4)	81.8%	+18.1	0.002
8 (ES→4)	82.2%	+18.6	0.002

6.2. General Reasoning Evaluation

A common concern with domain-specific fine-tuning is catastrophic forgetting of general capabilities. To assess whether our SFT and GRPO pipeline preserves or improves general reasoning, we evaluate the base and fine-tuned models on standard mathematical reasoning benchmarks.

[TODO: Present before/after comparison on GSM8K, MATH, or other standard reasoning benchmarks. Discuss whether domain-specific SFT caused catastrophic forgetting or whether the chain-of-thought training transferred positively to general math reasoning.]

6.3. Analysis

6.3.1. Dataset Size Scaling

Figure **??** plots SFT improvement as a function of training set size. The relationship is clearly non-linear: improvements are marginal below 300 examples (+2.0–2.3%), begin to emerge at 501 examples (+6.2%, $p = 0.063$), and become large and statistically significant at 866 examples (+18.6%, $p = 0.002$). This suggests a threshold effect around 500 examples, below which LoRA fine-tuning on this domain struggles to overcome the regularization imposed by the frozen base weights.

[TODO: Add scaling curve figure.]

The question flip ratio (Table 9) is particularly informative: at 215 samples, the model trades nearly as many correct answers as it gains (1.3:1), while at 866 samples, gains overwhelm regressions (4.6:1). This indicates that larger augmented datasets not only improve average performance but also make the improvement more *robust*, with fewer catastrophic regressions on individual questions.

6.3.2. Epoch Convergence

Table 10 shows SFT improvement as a function of training epochs on the 866-question dataset (10-fold CV). Performance increases rapidly from 3 to 4 epochs, then plateaus. Early stopping with patience 2 independently selects epoch 4 in both Exp. 22 (max 6) and Exp. 24 (max 8), confirming this as a stable convergence point.

6.3.3. Error Analysis

[TODO: Qualitative analysis of what the fine-tuned model gets right vs. wrong. Breakdown by reliability engineering sub-topic (Weibull, Bayesian, system reliability, etc.). Common failure modes: multi-step Bayesian problems, competing risks, edge cases with truncated distributions.]

7. Conclusion

We presented a complete pipeline for adapting a general-purpose LLM to reliability engineering, a technical domain with no existing training datasets or benchmarks. Starting from 56 textbook-extracted seed questions, our adapted self-instruct methodology with cross-model answer verification and paraphrase augmentation scales the dataset to 866 verified examples. Supervised fine-tuning of Qwen3-8B yields a statistically significant +18.6% accuracy improvement ($p = 0.002$) under 10-fold cross-validation, placing our 8B-parameter model between frontier models o3-mini and Gemini 3.1 Pro on domain-specific tasks.

Our analysis reveals two key findings for the synthetic data generation community. First, the *self-instruct ceiling effect*: LLM generators create questions within their own capability envelope, making same-model answer consistency a trivially satisfied check rather than a meaningful quality filter. Second, paraphrase augmentation—creating surface-level rephrases while preserving mathematical content—is far more effective than generating harder synthetic questions, consistent with MetaMath [24] and suggesting that diversity of expression matters more than difficulty in small-data regimes.

[TODO: Add GRPO results and discussion once available.]

[TODO: Add discussion of general reasoning benchmark results (catastrophic forgetting vs. positive transfer).]

Limitations. Our evaluation is limited to numeric questions with automated answer comparison; formula derivation and conceptual questions require human evaluation. The self-instruct ceiling effect means our synthetic data may not cover the hardest problems in reliability engineering. Additionally, our cross-validation evaluates on in-distribution paraphrased questions; fully out-of-distribution evaluation on unseen problem types is ongoing.

Future work. Promising directions include grounding data generation in textbook source material (following SearchInstruct [8]), extending to non-numeric question types, evaluating on other engineering domains to test the generality of our pipeline, and scaling GRPO training with the full augmented dataset.

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A. Complete Experiment Log

Table 11 lists all 26 experiments with full configurations and results. All experiments use 4-bit QLoRA via Unsloth on NVIDIA A100-SXM4-40GB GPUs. Evaluation uses greedy decoding (`do_sample=False`) from Exp. 7 onwards; earlier experiments used stochastic sampling (marked with †). Bold rows indicate best results per dataset size. ES = early stopping (patience=2); the notation "8(ES→4)" means max 8 epochs with early stopping selecting epoch 4.

[TODO: ADD YOUR EXPERIMENTS ELORA IN THIS TABLE OR A SECOND]

Notes: Exp. 1 uses Llama 3.1 8B; all others use Qwen3-8B. Exp. 5 uses LoRA $r = 8/\alpha = 16$; all others use $r = 16/\alpha = 32$. Exp. 10 timed out after 14 hours. Exp. 13 crashed due to Triton kernel incompatibility. Exp. 14 completed only 4 of 5 folds (missing adapter for fold 4). Accuracy values are percentages; Δ is the absolute difference (FT − Baseline). W→R and R→W are question flip counts aggregated across all folds.

B. Key Prompts

Below are selected prompts used in our data pipeline. Minor formatting adjustments have been made for readability.

B.1. Synthetic Question Generation

Used to generate new reliability engineering problems from seed examples (Section 3.2).

> *You are an expert in reliability engineering, creating practice problems for students.*
>
> *I will show you {num_examples} example problems. Your task is to:*
>
> 1. *Understand the style, difficulty, and topics of these examples*
> 2. *Create {num_to_generate} new, original problems that are similar in style and difficulty*
> 3. *Make questions that are educational and challenging*
> 4. *Provide complete solutions with step-by-step reasoning*
>
> *Important: Create different questions (don't just change numbers in the examples). Use similar concepts but new scenarios. Include all necessary data in the question (self-contained). Show clear reasoning steps. Provide specific numerical answers when appropriate.*
>
> *[3 randomly sampled seed examples shown]*
>
> *For each problem, provide:*
> *QUESTION: [Complete problem statement]*
> *REASONING: [Step-by-step solution]*
> *ANSWER: [Final answer(s)]*

B.2. Paraphrase Augmentation

Used to rephrase existing questions while preserving mathematical content (Section 3.4). The prompt includes two manually curated few-shot examples (omitted for brevity).

> *You are a technical writer specializing in reliability engineering. Your task is to rephrase a question and its solution reasoning while preserving their exact mathematical meaning.*
>
> *Strict rules:*
>
> 1. *All numerical values, failure rates, probabilities, time periods, and parameters must remain exactly the same*
> 2. *All formulas, equations, and mathematical relationships must be preserved*
> 3. *The correct answer must be identical*
> 4. *Change the wording, sentence structure, and phrasing to create a distinct version*
> 5. *Do not add information, assumptions, or conditions not present in the original*
>
> *What to change: Synonyms for non-technical words (e.g., "compute" → "determine"), sentence structure, scenario framing (e.g., "plant" → "factory"), transition words.*
>
> *What must not change: Any number, rate, probability, or parameter value. Variable names in formulas. The mathematical operations and formulas used. The final answer.*
>
> *[2 few-shot examples shown]*
>
> *Respond with a JSON object:* `{"question": "...", "reasoning": "..."}`

B.3. Paraphrase Verification

Used by GPT-5.4 to verify that a rephrase preserves mathematical meaning (Section 3.4).

> *You are verifying whether a rephrased reliability engineering question preserves the exact mathematical meaning of the original. Compare the original and rephrased versions. Check all of the following:*
>
> 1. *Numerical values: Are all numbers, rates, probabilities exactly the same?*
> 2. *Mathematical meaning: Do both questions ask for the exact same quantity?*
> 3. *Constraints: Are all conditions and given information preserved?*
> 4. *Solution: Would the exact same calculation steps produce the correct answer?*
> 5. *Reasoning: Are all formulas and intermediate values identical?*
>
> *Respond with a JSON object:*
> `{"values_preserved": bool,`
> `"meaning_preserved": bool,`
> `"constraints_preserved": bool,`

Table 11. Complete experiment log. All models are Qwen3-8B (4-bit) except Exp. 1 (Llama 3.1 8B). LoRA config is r/α with dropout 0.05 unless noted. † = stochastic eval (inflated). ‡ = thinking enabled (mismatch).

Exp	Dataset	N	Folds	Ep.	LR	NEF	Method	Baseline	FT	Δ	W→R	R→W	p
Round 1: Model Selection													
1	cleaned (all)	256	5	3	2e-4	–	SFT†	37.5	39.8	+2.4	35	29	0.25
2	cleaned (num)	215	5	4	5e-5	–	SFT†‡	67.9	67.0	−0.9	25	27	0.88
Round 2: Hyperparameter Search (stochastic eval)													
3	cleaned (num)	215	5	3	2e-4	5	SFT†	67.9	74.9	+7.0†	31	16	0.13
4	cleaned (num)	215	5	3	2e-4	10	SFT†	70.7	72.1	+1.4	21	18	1.0
5	cleaned (num)	215	5	3	2e-4	5	SFT†	67.9	69.3	+1.4	24	21	0.88
6	cleaned (num)	215	5	3	1e-4	5	SFT†	68.8	68.4	−0.5	20	21	1.0
Round 3: Deterministic Evaluation (greedy decoding)													
7	v2 (num)	280	5	3	2e-4	5	SFT	64.0	65.2	+1.3	21	21	0.88
8	v2 (num)	280	5	5	2e-4	5	SFT	64.3	65.7	+1.4	30	19	0.88
9	cleaned (num)	215	5	5	2e-4	5	SFT†	70.7	72.6	+1.9	21	17	0.50
11	cleaned (num)	215	5	3	2e-4	5	SFT	70.7	71.6	+0.9	22	20	0.88
12	**cleaned (num)**	**215**	**5**	**4**	**2e-4**	**5**	**SFT**	**70.7**	**73.0**	**+2.3**	**23**	**18**	**0.50**
16	**v2 (num)**	**280**	**5**	**4**	**2e-4**	**5**	**SFT**	**71.0**	**73.0**	**+2.0**	**19**	**15**	**0.69**
17	cleaned (num)	215	5	4	2e-4	7	SFT	71.0	69.8	−1.3	19	18	0.38
Round 4: Alternative Methods													
10	v2 (num)	280	5	DPO	5e-6	–	DPO	64.3	–	–	–	–	–
14	cleaned (num)	215	5	DPO	5e-6	–	DPO	69.8	68.0	−1.7	4	7	0.63
13	cleaned (num)	215	5	SFT+GRPO	2e-4	5	GRPO	70.7	–	–	–	–	–
15	cleaned (num)	215	5	SFT+GRPO	2e-4	5	GRPO	70.7	71.2	+0.5	18	17	0.88
Round 5: Paraphrase Augmentation (5-fold CV)													
18	**v3 (num)**	**501**	**5**	**4**	**2e-4**	**5**	**SFT**	**59.1**	**65.3**	**+6.2**	**74**	**43**	**0.063**
19	v3 (num)	501	5	8(ES→4)	1.5e-4	5	SFT	59.1	64.5	+5.4	83	56	0.063
20	v3 (num)	501	5	6	1e-4	5	SFT	59.1	64.9	+5.8	78	49	0.063
Round 6: Scaled Augmentation (10-fold CV)													
23	v4 (num)	866	10	3	2e-4	5	SFT	63.6	75.1	+11.4	171	72	**0.002**
21	v4 (num)	866	10	4	2e-4	5	SFT	63.6	79.3	+15.7	194	58	**0.002**
22	v4 (num)	866	10	6(ES→4)	2e-4	5	SFT	63.6	81.8	+18.1	208	51	**0.002**
24	**v4 (num)**	**866**	**10**	**8(ES→4)**	**2e-4**	**5**	**SFT**	**63.6**	**82.2**	**+18.6**	**206**	**45**	**0.002**
Round 7: 600-Sample Subset Validation (10-fold CV)													
25	**v4 sub (num)**	**600**	**10**	**5**	**2e-4**	**5**	**SFT**	**68.0**	**79.7**	**+11.7**	**108**	**38**	**0.002**
26	v4 sub (num)	600	10	5(ES)	2e-4	5	SFT	68.0	78.8	+10.8	100	35	**0.002**

```
"solution_equivalent": bool,
"reasoning_equivalent": bool,
"overall_valid": bool, "issues":
"..."}
```

B.4. SFT System Prompt

Used as the system message during both training and evaluation (Section 4.2).

You are an expert in reliability engineering, probability theory, and system analysis. Provide clear, accurate answers with step-by-step reasoning. At the end, clearly state your final answer after 'Final Answer:'.